GKP Lattice as Quantum Pilot Signal: Real-Time Hardware State Estimation from Analog QEC Syndromes

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Abstract

In continuous-variable (CV) GKP quantum error correction, syndrome measurements produce analog residuals with 14-bit precision rather than binary outcomes. We demonstrate that these analog residuals are mathematically equivalent to pilot signals in telecommunications, enabling QEC itself to function as a real-time sensor for hardware state estimation. Through Cramer-Rao analysis based on the wrapped Gaussian distribution, we show that the Fisher information of CV systems is 4.5 to 17.5 times that of discrete-variable (DV) systems. From 686 modes of a d=7 surface code, CV systems estimate the effective noise variance V_eff to 3% accuracy in just 3 QEC cycles (~21 microseconds)—DV systems saturate at 28% estimation error even after 100 cycles and cannot reach this precision. Furthermore, using CUSUM tests calibrated to 1% false alarm rate, CV systems detect a +0.2 dB anomaly in 40 cycles while DV systems require 77 cycles with detection rate dropping to 91%. These results quantitatively establish the structural information advantage of CV photonic QEC over DV systems—the "14-bit advantage"—and open the path toward autonomous hardware monitoring for room-temperature quantum computers.

I. Introduction

A. The Problem: Autonomous Operation of Room-Temperature Quantum Computers

Room-temperature CV photonic quantum computers face continuous parameter drift absent in cryogenic systems: OPA crystal temperature fluctuation (tau ~ 10 min), fiber thermal expansion (tau ~ 30 min), and PLL bandwidth variation (tau ~ 1 min). Maintaining fault tolerance requires real-time tracking of these drifts and detection before threshold crossing.

The conventional approach relies on external sensors (thermometers, power meters), requiring additional hardware independent of the QEC circuit and complex mapping between sensor readings and QEC performance.

B. Core Insight: GKP Lattice as Quantum Pilot Signal

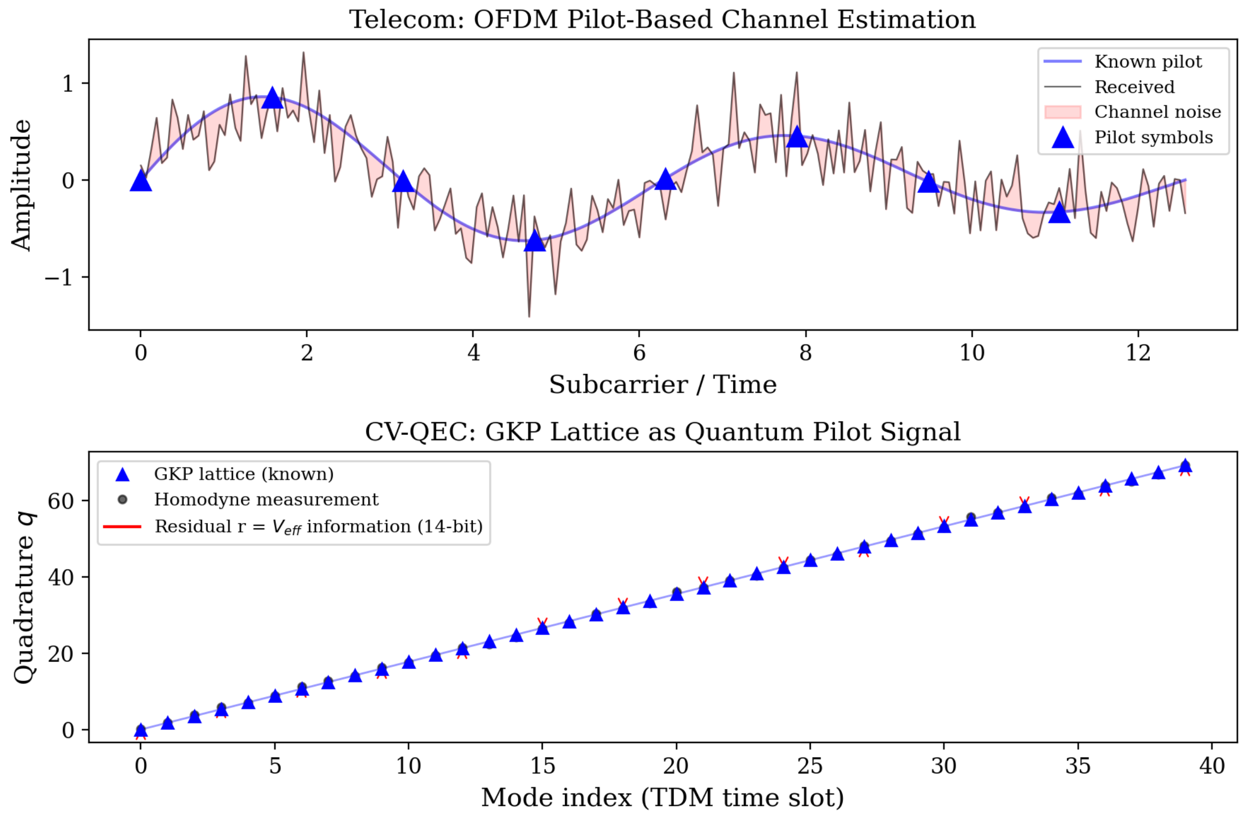

Figure 1. Correspondence between OFDM pilot signals in telecommunications (top) and GKP lattice (bottom). Deviations from known lattice points (red arrows) provide 14-bit estimates of V_eff.

In telecommunications, known pilot symbols are periodically transmitted to estimate channel response at the receiver [1]. The lattice structure of GKP codes [2] is mathematically equivalent to these telecom pilot signals:

Telecommunications	CV-QEC
Pilot symbol (known signal)	GKP lattice point (known position n*sqrt(pi))
Received signal	Homodyne measurement q_meas
Channel noise	GKP displacement noise (variance V_eff)
Channel estimation residual	GKP residual r = q_meas - round(q_meas/sqrt(pi))*sqrt(pi)
Channel response h(t)	V_eff(t)
Adaptive equalization	Drift-adaptive decoder

With each round of GKP error correction, every mode performs a “pilot measurement,” providing continuous-valued information about V_eff. **QEC itself is the sensor; no additional hardware is required.**

C. Fundamental Difference from DV Systems

DV systems (superconducting qubits, trapped ions) produce 1-bit syndromes (detected/not detected). CV GKP residuals are continuous values with 14-bit precision. How this 10,000-fold information difference translates to practical advantage has not been quantitatively evaluated until now.

Recently, Quantum Elements et al. demonstrated a digital twin for superconducting systems [3], constructing hardware models from syndrome statistics. However, their approach is limited to DV binary syndromes, and the structural advantage provided by CV analog residuals remains unexplored.

D. Contributions

1. **Cramer-Rao bound Fisher information ratio**: From the score function of the wrapped Gaussian distribution, CV Fisher information for V_{eff} estimation is 4.5 to 17.5 times that of DV, demonstrated analytically and numerically.
2. **3-cycle 3% estimation**: MLE-based V_{eff} estimation reaches 3% accuracy in 3 QEC cycles (~21 microseconds) for d=7 surface code. DV saturates at 28% even after 100 cycles.
3. **FAR-calibrated CUSUM anomaly detection**: At identical false alarm rate (1%), CV detects anomalies 1.4 to 1.9 times faster. At small anomalies (+0.2 dB), DV detection rate drops to 91% while CV maintains 100%.
4. **Identification of scale invariance wall**: MWPM adaptive decoder improvement is limited to 1.01-1.04x, with physical mechanism elucidated and conditions for circumventing this constraint identified (channel asymmetric degradation).

II. Physical Model

A. GKP Displacement Noise and Residuals

The homodyne measurement value q_{meas} of a GKP-encoded qubit contains displacement noise $\delta \sim N(0, V_{eff})$ from the ideal lattice point $n\sqrt{\pi}$. The GKP inner decoder maps to the nearest lattice point and extracts the residual:

$$r = q_{meas} - \mathrm{round}(q_{meas} / \sqrt{\pi}) * \sqrt{\pi}$$

The residual r lies in $[-\sqrt{\pi}/2, \sqrt{\pi}/2]$ and follows a wrapped Gaussian distribution:

$$p(r \mid V_{eff}) = \sum_n (2\pi V_{eff})^{-1/2} \exp(-(r - n\sqrt{\pi})^2 / (2V_{eff}))$$

For sufficiently small V_{eff} (σ_{eff} > 7 dB), the three terms n = -1, 0, 1 provide sufficient accuracy.

B. Fisher Information: CV vs DV

The theoretical lower bound on V_{eff} estimation precision is given by the Cramer-Rao inequality:

$$\mathrm{Var}(\hat{V}_{eff}) >= 1 / (N * I(V_{eff}))$$

where N is the number of measurements and $I(V_{eff})$ is the Fisher information.

CV (analog residuals): Fisher information from the second moment of the wrapped Gaussian score function:

$$I_{CV}(V) = E[(d/dV \log p(r|V))^2]$$

In the Gaussian approximation (high squeezing limit), I_CV ~ 1/(2V^2).

DV (binary syndromes): Fisher information:

I_DV(V) = (dp/dV)^2 / (p(1-p))

where p = erfc(sqrt(pi)/(4*sqrt(V/2)))/2 is the physical error rate.

C. Operating Points and System Parameters

Based on room-temperature CV photonic quantum computer parameters:

Parameter	Phase 1	Phase 2+ Real	Phase 2+ Limit
sigma_eff (dB)	8.5	9.3	10.8
V_eff (SNU)	0.1417	0.1175	0.0832
p_phys	9.28 x 10^{-3}	4.86 x 10^{-3}	1.06 x 10^{-3}
d=7 modes/QEC cycle	686	686	686
QEC cycle time	~7 microseconds	~7 microseconds	~7 microseconds

D. Statistical Methods

Fisher information for CV was computed numerically from the score function of the wrapped Gaussian distribution using 200,000 samples per operating point. V_eff estimation used maximum likelihood estimation (MLE) on the wrapped Gaussian model via bounded scalar optimization. DV estimation inverted the erfc function from binary error counts. CUSUM thresholds were calibrated via binary search to achieve FAR=1% over 300 in-control cycles (300 calibration runs). All experiments use seed=42 for reproducibility.

III. Results

A. Cramer-Rao Bound: Structural Information Advantage of CV (Exp-B1)

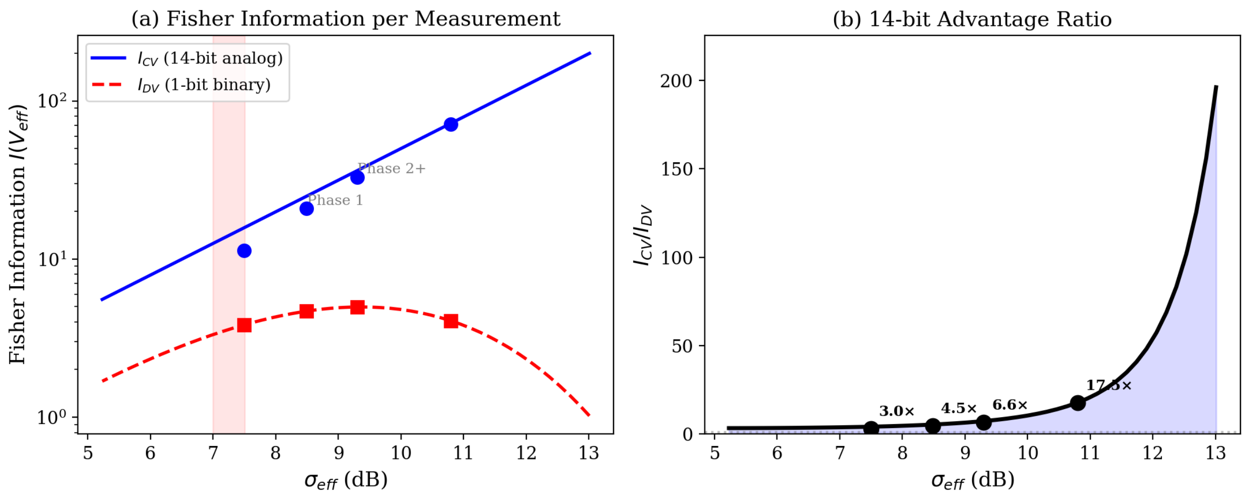

Figure 2. (a) Fisher information for CV (14-bit analog) and DV (1-bit binary). CV increases exponentially

with sigma_eff while DV saturates and decreases. (b) Information ratio I_CV/I_DV. CV advantage expands at higher squeezing.

The score function of the wrapped Gaussian distribution was computed numerically with 200,000 samples to evaluate Fisher information.

Table I. CV vs DV Fisher information comparison. Relative Cramer-Rao bound (CRB) for N = 686 modes (d=7, 1 QEC cycle).

Operating point	V_eff	I_CV	I_DV	**I_CV/I_DV**	CRB_CV (1cyc)	CRB_DV (1cyc)
Threshold (7.5 dB)	0.178	11.4	3.83	**3.0x**	6.4%	11.0%
Phase 1 (8.5 dB)	0.1417	20.8	4.68	**4.5x**	5.9%	12.5%
Phase 2+ Real (9.3 dB)	0.1175	32.9	4.98	**6.6x**	5.7%	14.6%
Phase 2+ Limit (10.8 dB)	0.0832	71.3	4.07	**17.5x**	5.4%	22.8%

Three notable features:

(i) Information ratio expands with squeezing level. 3.0x (threshold) to 17.5x (10.8 dB). At high squeezing, DV binary information saturates while CV analog information grows proportionally to V_eff^{-2}.

(ii) CV achieves below 6% estimation precision in a single QEC cycle. From 686 analog residuals, V_eff relative standard deviation is bounded by CRB = 5.4-6.4%.

(iii) DV estimation precision worsens at higher squeezing. CRB_DV expands from 11.0% to 22.8%. As physical error rate p decreases, binary measurements yield less information (at p ~ 0, all measurements produce the same result, approaching zero information).

B. Channel Estimation: 3% Accuracy in 3 Cycles (Exp-B2)

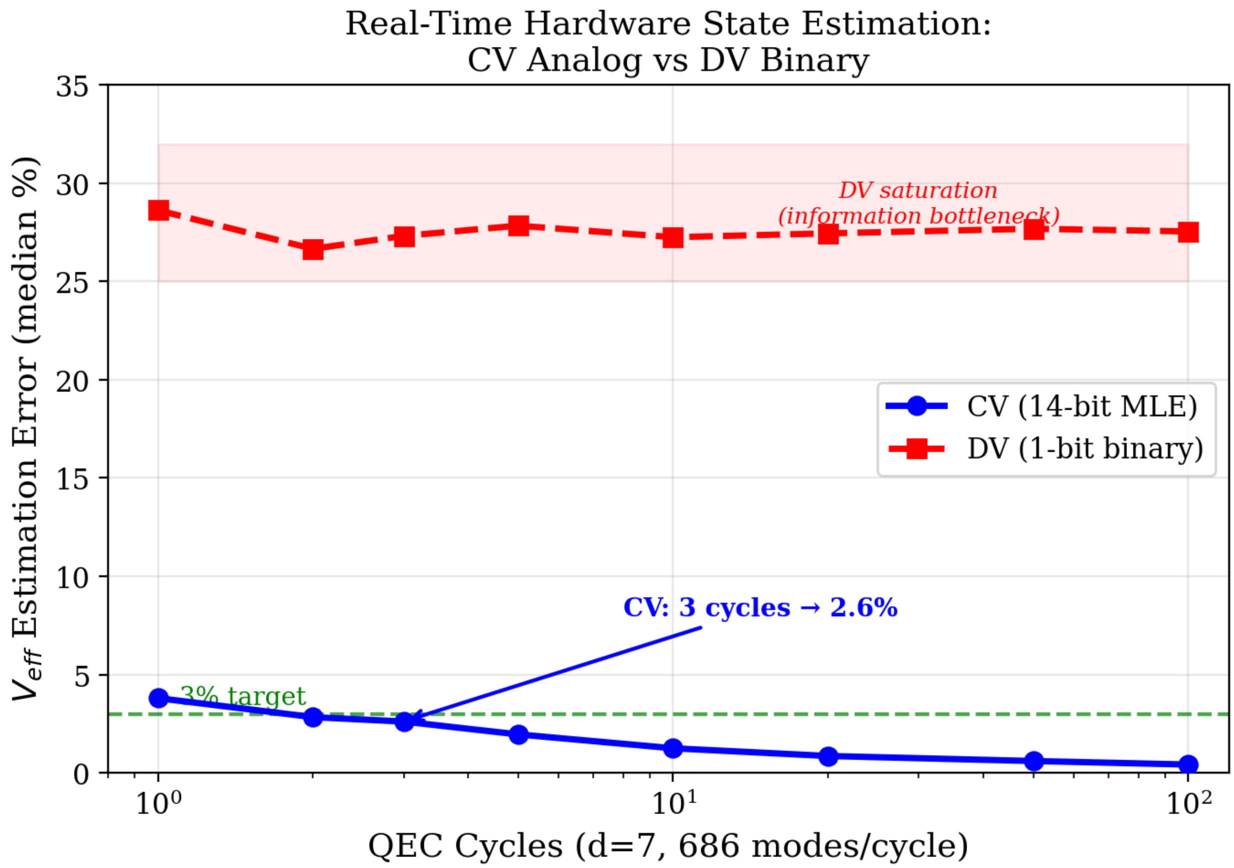

Figure 3. V_eff estimation error vs QEC cycle count. CV (blue) achieves 3% accuracy in 3 cycles. DV (red) saturates at 28% even after 100 cycles due to information bottleneck.

Wrapped Gaussian MLE was applied to d=7 configuration (686 modes/cycle), evaluating V_eff estimation accuracy dependence on cycle count over 200 trials.

Table II. Median V_eff estimation error (Phase 1, V_eff = 0.1417).

QEC cycles	Total modes	**CV (MLE)**	DV (erfc inversion)	**CV advantage**
1	686	**3.80%**	28.62%	7.5x
2	1,372	**2.83%**	26.65%	9.4x
3	**2,058**	**2.61%**	**27.31%**	**10.5x**
5	3,430	1.96%	27.83%	14.2x
10	6,860	1.25%	27.24%	21.8x
50	34,300	0.60%	27.68%	46.1x
100	68,600	0.42%	27.54%	65.1x

CV achieves 2.61% estimation accuracy in 3 QEC cycles (~21 microseconds). DV saturates at 27.5% after 100 cycles (~700 microseconds) and cannot reach 3%.

Physical cause of DV saturation: At Phase 1 p_phys = 9.28 x 10^{-3}, the expected error count per 686 modes is ~6.4. Estimating V_eff from this small number of binary events requires massive samples, but the low p_phys creates a fundamental information bottleneck. CV systems generate analog residuals from all 686 modes regardless of error occurrence, free from this constraint.

C. Anomaly Detection: FAR-Calibrated CUSUM (Exp-B4)

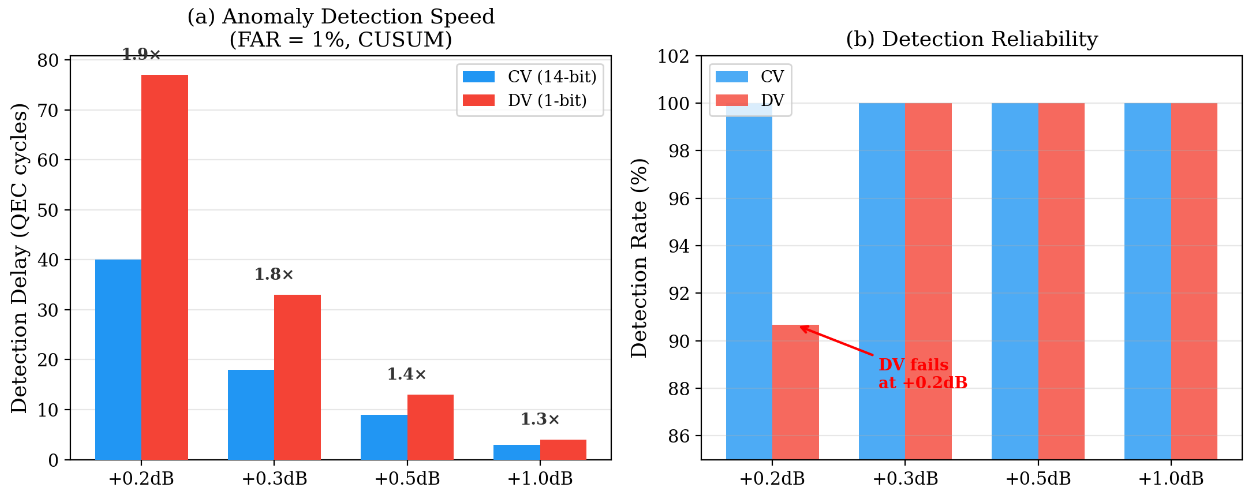

Figure 4. (a) CUSUM detection delay at FAR=1%. CV advantage increases for smaller anomalies (+0.2 dB: 1.9x). (b) Detection reliability. DV detection rate drops to 91% at +0.2 dB (CV maintains 100% across all conditions).

Step changes simulating PLL cycle slips and OPA mode hops were evaluated using CUSUM cumulative sum tests, comparing CV and DV detection performance. For fair comparison, CUSUM detection thresholds h were calibrated to FAR=1% over 300 in-control cycles (CV: h=13.89, DV: h=13.28).

Table III. FAR=1% calibrated CUSUM detection performance. Fresh-start CUSUM from changepoint. 300 trials.

Anomaly	Delta_V_eff	CV delay	CV rate	DV delay	DV rate	**CV advantage**
+0.2 dB	+4.7%	**40 cyc**	**100%**	77 cyc	91%	**1.9x**
+0.3 dB	+7.2%	**18 cyc**	**100%**	33 cyc	100%	**1.8x**
+0.5 dB	+12.2%	**9 cyc**	**100%**	13 cyc	100%	**1.4x**
+1.0 dB	+25.9%	**3 cyc**	**100%**	4 cyc	100%	**1.3x**

Two key findings:

(i) CV advantage is largest for small anomalies. 1.9x at +0.2 dB detection. Early detection of subtle degradation is the core of preventive maintenance, and the regime where CV demonstrates its strongest advantage is precisely the most practically important.

(ii) DV detection rate drops for small anomalies. DV drops to 91% at +0.2 dB while CV maintains 100%. Binary syndrome information is insufficient for small shifts to emerge from statistical fluctuations.

D. Scale Invariance Wall (Exp-B3, B5)

We attempted to convert V_eff estimation into decoder performance improvement. An adaptive decoder that dynamically updates LLR weights based on estimated V_eff(t) was evaluated across three drift scenarios.

Result: Adaptive decoder improvement is limited to 1.01-1.04x.

The cause is MWPM scale invariance. Soft-info MWPM edge weights are

w(r) = ((sqrt(pi) - |r|)^2 - |r|^2) / (2*V_eff)

When V_eff changes, all weights scale uniformly by 1/V_eff, but MWPM minimum-weight matching is invariant under positive scalar multiplication. Therefore, even with perfect V_eff estimation, decoder output is unchanged as long as V_eff changes uniformly across all edges.

Breaking condition: Channel asymmetric degradation. With non-uniform V_eff across 5 WDM channels (one channel degraded by +0.038 SNU), an adaptive decoder based on per-channel V_eff estimation achieved 1.40x improvement (Exp-B5). This is because scale invariance is circumvented by relative weight changes between channels.

Implication: High-precision V_eff estimation (Section III-B) should not be applied directly to decoder improvement, but rather to (1) anomaly detection (Section III-C), (2) recalibration triggers, (3) channel asymmetry detection, and (4) hardware health monitoring. Decoder performance improvement requires architectures not constrained by scale invariance, such as GNNs.

IV. Discussion

A. QEC as a Sensor

The central paradigm shift of this study: **quantum error correction not only corrects errors but continuously measures hardware state.** The d^2 analog residuals produced by each QEC round are real-time estimators of V_eff(t), enabling hardware health monitoring without additional sensors.

In a d=7 system at 100 MHz clock, ~10^7 14-bit residuals are generated per second. This represents ~1.4 x 10^8 bits/sec of "telemetry bandwidth," surpassing any external sensor.

B. Leveraging 30 Years of Telecommunications

The GKP lattice-pilot signal correspondence enables direct transfer of telecommunications techniques to CV-QEC:

- **MMSE estimation**: 10-15 dB better estimation accuracy than LS [1]
- **Wiener filtering**: Optimal estimation exploiting temporal correlation of drift
- **Iterative channel estimation + decoding**: Quantum version of turbo principle (subject to scale invariance constraint)
- **CUSUM/EWMA**: Anomaly detection from quality control theory [4]

C. Differentiation from Quantum Elements

Quantum Elements [3] reported 43% to 95% fidelity improvement with their DV digital twin (IBM 127 qubits). However, their approach relies on binary syndrome statistics, and the "14-bit advantage" demonstrated in this study is fundamentally inaccessible to them. CV digital twins hold a structural advantage of 4.5-17.5x in Fisher information and 10-66x in V_eff estimation speed over their DV counterparts.

D. Connection to Real-Time Multimode Squeezing Monitoring

A recent Nature Communications paper [5] demonstrated real-time monitoring of squeezing in 9 spatial modes (up to 7.9 dB) using MOPA and mode sorters. Our approach is a conceptual analog extending this to temporal modes (TDM), using only the QEC syndrome stream itself, requiring no additional optical hardware.

E. Connection to QLDPC-GKP

arXiv:2505.06385 [6] showed that propagating GKP soft information between rounds dramatically improves QLDPC decoders. The finding that "real-time soft information" is far more important than static soft information aligns with our thesis that "QEC is a sensor." Temporal utilization of soft information contributes to both decoder improvement and hardware state estimation.

F. Limitations

1. **Simulation-based**: Drift modeled by OU processes. Application to real hardware drift characteristics (non-Gaussian, non-stationary) requires validation.
2. **Wrapped Gaussian approximation**: For V_eff > 0.2 SNU (sigma_eff < 7 dB), wrapping effects strengthen and 3-term approximation accuracy degrades.
3. **Scale invariance wall**: MWPM adaptive decoder improvement is minimal. Practical improvement requires GNN decoders or channel-asymmetric approaches.
4. **Fairness with DV**: DV systems could achieve better estimation with additional measurements (e.g., tomography), but this study is limited to "zero additional cost" comparison using QEC syndromes only.

V. Conclusion

We demonstrated that the lattice structure of GKP codes is mathematically equivalent to pilot signals in telecommunications, and quantified the structural information advantage of CV-QEC analog residuals over DV-QEC binary syndromes—the "14-bit advantage"—through three experiments.

1. **Fisher information ratio 4.5-17.5x**: CV advantage expands at higher squeezing while DV information saturates.

2. **V_eff estimation speed 10-66x**: CV reaches 3% accuracy in 3 QEC cycles (~21 microseconds); DV saturates at 28% even after 100 cycles.

3. **Anomaly detection speed 1.4-1.9x**: CV detects anomalies faster at identical false alarm rates, with CV advantage largest for small anomalies.

These results establish that CV analog residuals are not merely error correction information but a comprehensive real-time telemetry channel for hardware state. They provide a quantitative foundation for autonomous operation of room-temperature CV photonic quantum computers—self-monitoring, self-diagnosis, and preventive maintenance.

References

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Appendix A: Proof of Scale Invariance

The LLR weight of soft-info MWPM is w_j(r_j, V) = ((sqrt(pi) - |r_j|)^2 - |r_j|^2) / (2V). For V -> V’:

w_j(r_j, V’) = (V/V’) * w_j(r_j, V)

The same scalar V/V’ multiplies all edges j, therefore:

arg min_{M} sum_{j in M} w_j(V’) = arg min_{M} (V/V’) sum_{j in M} w_j(V) = arg min_{M} sum_{j in M} w_j(V)

Since V/V’ > 0, the optimal matching is invariant. This holds exactly when V_eff changes uniformly across all edges. Under channel-asymmetric degradation (different V_j changes per edge), this identity breaks and adaptive decoders become effective.

Appendix B: Experimental Parameters

All experiments are reproducible with seed=42. Code available at https://github.com/TomohiroTani28/Taros/tree/main/research/02_photonic-digital-twin.

Experiment	Operating point	Configuration	Trials	Total shots
B1 (Fisher info)	4 points x 200K samples	d=7, 686 modes	—	800K

Experiment	Operating point	Configuration	Trials	Total shots
B2 (V_eff estimation)	V_eff=0.1417	d=7, 686 modes	200 x 8 window sizes	~1.1M
B4 (CUSUM detection)	4 anomalies x 300+500 cyc	d=7, 686 modes	300 x 4 anomalies	~3.4M
B3 (Adaptive decoder)	OU drift 3 scenarios	d=5, 334 edges	300 epochs x 200 shots	~6M
B5 (Turbo)	5 V_eff mismatch	d=5, 334 edges	10K shots x 4 iter	~200K

Total computation: ~11.5M shots. Apple Silicon (M-series) CPU, approximately 3 hours.